1.What is Git?

● Define Git as a version control system

● Explain its purpose (tracking changes, collaboration, etc.)

Git is a distributed version control system that allows developers to record, manage, and track changes in their code or files.

Purpose of Git

Tracking Changes

Git keeps a history of every change made to your files.

You can go back to any previous version if something breaks.

Collaboration

Multiple people can work on the same project at the same time.

Git helps merge changes from different team members without conflicts.

Version Management

You can create different versions (called branches) of your project.

Useful for testing new features without affecting the main code.

Backup & Safety

Your code is safely stored and can be recovered anytime.

Mistakes can be undone easily.

Distributed System

Every user has a full copy of the project history.

Work can continue even without internet

2. What is Version Control?

● Explain the concept of version control

● Importance in software development

Version Control is a system that helps you track, manage, and store changes made to files (especially code) over time.

Concept of Version Control

The main idea behind version control is:

Tracking Changes: Every modification in the file is recorded

Version History: You can view previous versions

Branching: Work on different features separately without affecting the main code

Merging: Combine different changes into one final version

Collaboration: Multiple people can work on the same project without conflict

Importance in Software Development

Version control is very important because:

1. Prevents Data Loss

If something breaks, you can restore an older version

2. Enables Teamwork

Many developers can work together on the same project

3. Tracks History

You can see what changes were made and why

4. Supports Experimentation

Developers can try new ideas without affecting the main project

5. Improves Productivity

Saves time and avoids confusion in large projects

3. What is GitHub?

● Define GitHub as a platform for hosting repositories

● Mention its role in collaboration

GitHub is an online platform used to host and manage repositories (projects) that use version control.

Definition

GitHub is a platform where developers:

Upload their projects (repositories)

Track changes using Git

Access their code from anywhere

Role in Collaboration

GitHub plays a major role in teamwork and collaboration:

1. Multiple Developers Can Work Together

Many people can work on the same project at the same time

2. Pull Requests

Developers can suggest changes

Team members can review before accepting

3. Code Review & Discussion

You can comment on code

Discuss improvements directly in the project

4. Branching & Merging

Work on new features separately

Merge them safely into the main project

5. Centralized Project Storage

All project files are stored in one place (repository)

Easy to manage and share

4. Difference Between Git and GitHub

● Write at least 3–5 clear differences

Git and GitHub are related but different:

Git is a version control system, while GitHub is a platform for hosting Git repositories.

Git works on your local computer and can be used offline, but GitHub is an online service that requires internet.

Git is used to track changes and manage versions of code, whereas GitHub is used to store, share, and manage code online.

Git has limited collaboration features, but GitHub provides easy collaboration tools like pull requests and code reviews.

Git is a software tool, while GitHub is a web-based platform built on Git.

basic command and their description

git --version

Checks the installed Git version on

```
mdksm@Mohammed MINGW64 ~/OneDrive/Attachments/Desktop/gitbash
$ git --version
git version 2.52.0.windows.1
```

git config --global user.name "Your Name"

Sets the global username for Git commits.

git config --global user.email email@gmail.com

Sets the global email for Git commits.

git config user.name

Displays the configured Git username.

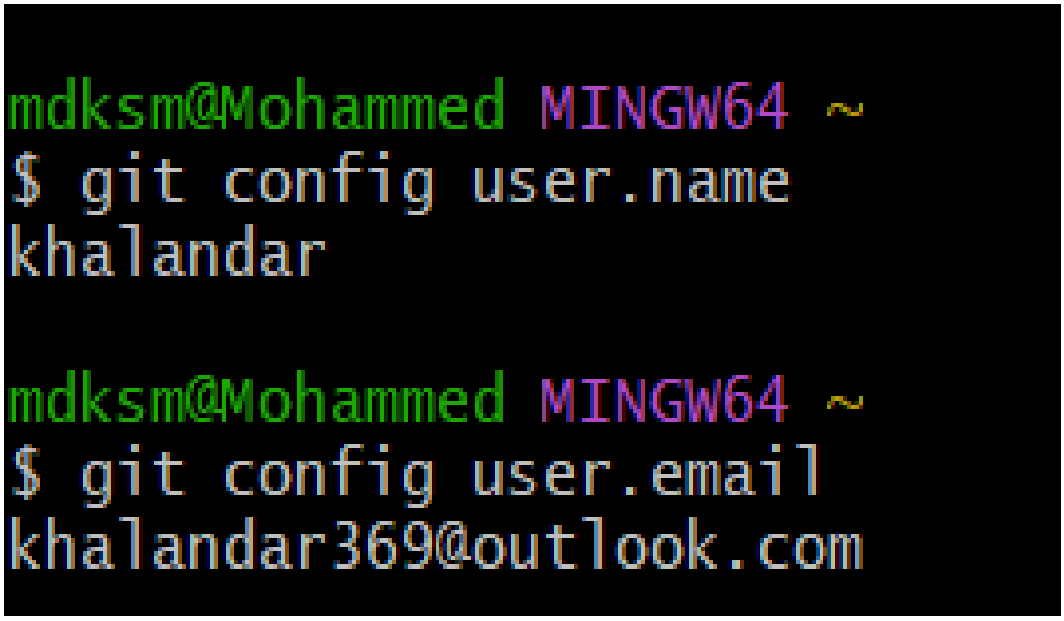
git config user.email

Displays the configured Git email.

git config --list

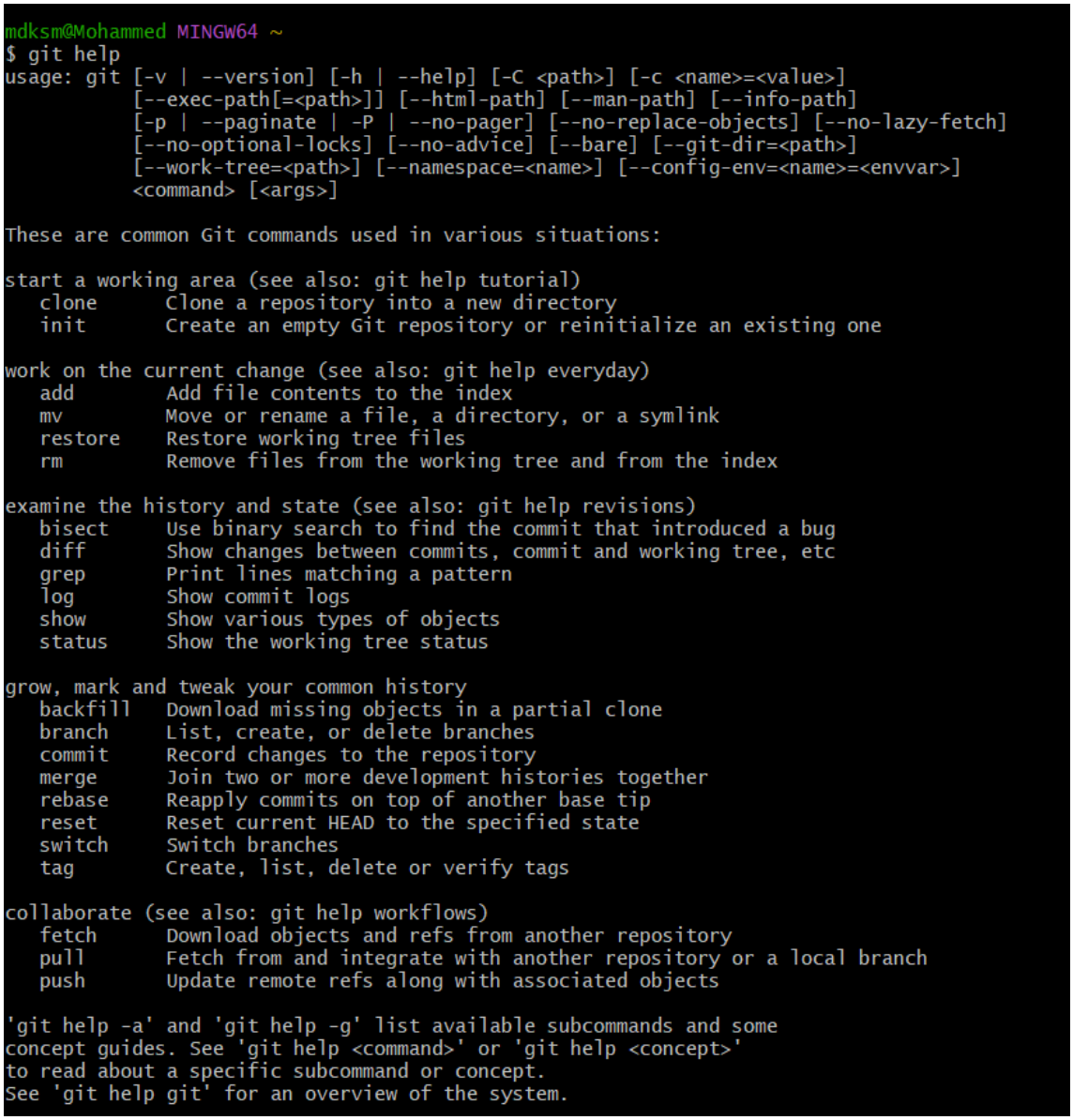
Shows all Git configuration settings.


MINGW64:/c/Users/mdksm
mdksm@Mohammed MINGW64 ~
$ git --version
git version 2.52.0.windows.1
mdksm@Mohammed MINGW64 ~
$ git config --global user.name "khalandar"
mdksm@Mohammed MINGW64 ~
$ git config --global user.email"khalandar369@outlook.com"
mdksm@Mohammed MINGW64 ~
$ git config --list
diff.astextplain.textconv=astextplain
filter.lfs.clean=git-lfs clean -- %f
filter.lfs.smudge=git-lfs smudge -- %f
filter.lfs.process=git-lfs filter-process
filter.lfs.required=true
http.sslbackend=schannel
core.autocrlf=true
core.fscache=true
core.symlinks=false
pull.rebase=false
credential.helper=manager
credential.https://dev.azure.com.usehttppath=true
init.defaultbranch=master
user.email=khalandar369@outlook.com
user.name=khalandar




git help

Displays help/manual for Git commands.




pwd

Shows the current working directory (present folder location).

cd

folder_name/location Changes the current directory to another folder.

Ls

Shows the current working directory (present folder location).

ls -l

Changes the current directory to another folder.

ls -a

Lists all files including hidden files (files starting with .)

ls -d */

Lists only directories (folders) in the current location.

```
mdksm@Mohammed MINGW64 ~/OneDrive/Music
$  cd C:/Users/mdksm/OneDrive/Music

mdksm@Mohammed MINGW64 ~/OneDrive/Music
$ ls
Playlists/  desktop.ini

mdksm@Mohammed MINGW64 ~/OneDrive/Music
$ ls -1
Playlists/
desktop.ini

mdksm@Mohammed MINGW64 ~/OneDrive/Music
$ ls -a
./  ../  Playlists/  desktop.ini

mdksm@Mohammed MINGW64 ~/OneDrive/Music
$ ls -d */
Playlists//
```


touch filename

Creates a new empty file.

touch file1 file2 file3

Creates multiple files at once. rm filename Deletes a file.

mkdir folder_name

Creates a new folder.

mkdir folder1 folder2 folder3

Creates multiple folders at once.

rmdir folder_name

Deletes an empty folder.

rm -r foldername

Deletes a folder and its contents (non-empty folder).

```
mdksm@Mohammed MINGW64 ~/OneDrive/Music
$  touch file1.txt

mdksm@Mohammed MINGW64 ~/OneDrive/Music
$  touch file2.txt file3.txt file4.txt file5.txt file6.txt file7.txt file8.txt file9.txt file10.txt

mdksm@Mohammed MINGW64 ~/OneDrive/Music
$  mkdir python1 python2 python3 python4 python5 python6 python7 python8 python9 python10
```

```
mdksm@Mohammed MINGW64 ~/OneDrive/Music
$ rm file1.txt file2.txt

mdksm@Mohammed MINGW64 ~/OneDrive/Music
$ ls
Playlists/  file10.txt  file4.txt  file6.txt  file8.txt  python1/  python2/  python4/  python6/  python8/
desktop.ini  file3.txt  file5.txt  file7.txt  file9.txt  python10/  python3/  python5/  python7/  python9/

mdksm@Mohammed MINGW64 ~/OneDrive/Music
$ rmdir python10 python5

mdksm@Mohammed MINGW64 ~/OneDrive/Music
$ ls
Playlists/  desktop.ini  file10.txt  file3.txt  file4.txt  file5.txt  file6.txt  file7.txt  file8.txt  file9.txt  python1/  python2/  python3/  python4/  python6/  python7/  python8/  python9/

mdksm@Mohammed MINGW64 ~/OneDrive/Music
$
```

cat filename

Displays the contents of a file in the terminal.

echo "text-content" > filename

Writes text to a file. If the file does not exist, it will be created.

echo "text" >> filename

Appends text to an existing file without deleting previous content.

```
mdksm@Mohammed MINGW64 ~/OneDrive/Attachments/Desktop/gitbash
$ echo "practicing git" > gitfile.txt

mdksm@Mohammed MINGW64 ~/OneDrive/Attachments/Desktop/gitbash
$ cat gitfile.txt
practicing git
```

```
mdksm@Mohammed MINGW64 ~/OneDrive/Attachments/Desktop/gitbash
$ echo "git commands2" >>gitfile.txt

mdksm@Mohammed MINGW64 ~/OneDrive/Attachments/Desktop/gitbash
$ cat gitfile.txt
practicing git
git commands2
```

cat > filename

Creates a file and allows us to type text directly in the terminal.

Steps:

cat > file.txt

Type your text here Press Ctrl + D to save and exit

cat >> filename

Adds text to an existing file without deleting previous content.

Steps:

cat >> file.txt

Type additional text Press Ctrl + D to save

```
mdksm@Mohammed MINGW64 ~/OneDrive/Attachments/Desktop/gitbash
$ cat >> gitfile.txt
{gitcommand for creating appending and deleting files using echo and cat}
mdksm@Mohammed MINGW64 ~/OneDrive/Attachments/Desktop/gitbash
$ cat gitfile.txt
git commands for creating appending and deleting files using echo and{gitgit command for creating appending and deleting files using echo and cat}
mdksm@Mohammed MINGW64 ~/OneDrive/Attachments/Desktop/gitbash
$
```

cat file1.txt > file2.txt

Copies the content of file1.txt into file2.txt.


nano file.txt

Opens the Nano text editor to create or edit a file.

Opens Steps in Nano Editor:

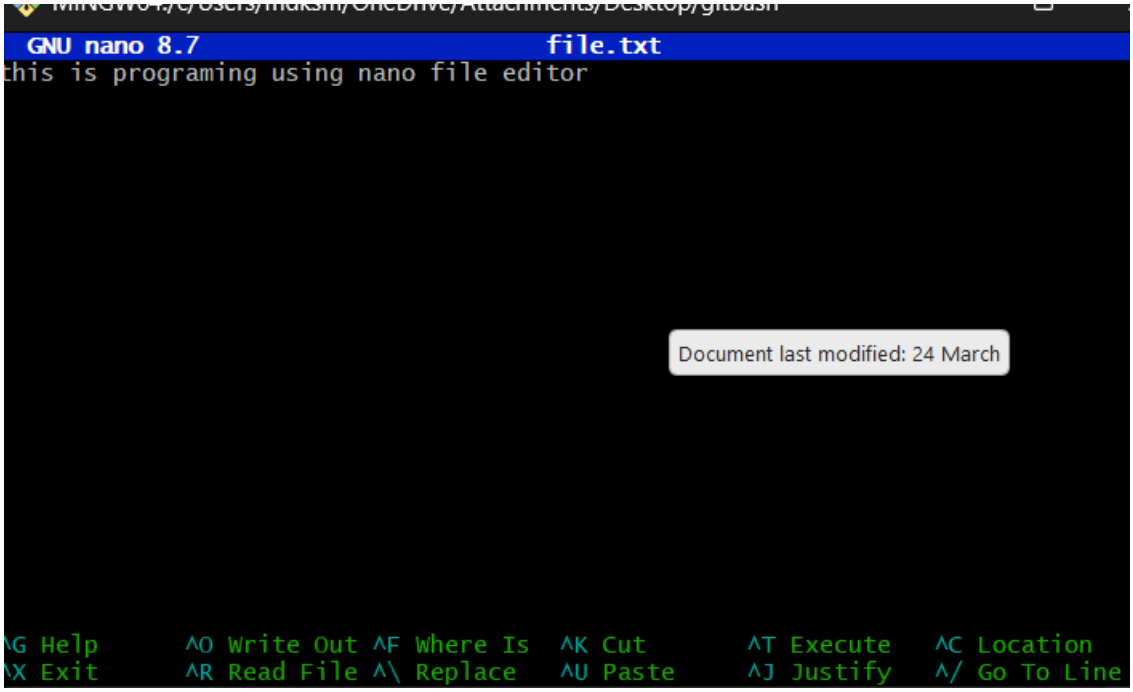
Save the file Ctrl + O

Exit the editor Ctrl + X

```
mdksm@Mohammed MINGW64 ~/OneDrive/Attachments/Desktop/gitbash
$ nano file.txt
```

```
mdksm@Mohammed MINGW64 ~/OneDrive/Attachments/Desktop/gitbash
$ ls
file.txt     gitfile2.txt   labfile.pdf.txt
gitfile.txt  khalandar.txt  mohammedkhalandar.txt

mdksm@Mohammed MINGW64 ~/OneDrive/Attachments/Desktop/gitbash
$ cat file.txt
this is programing using nano file editor
```




1. Check the default editor via

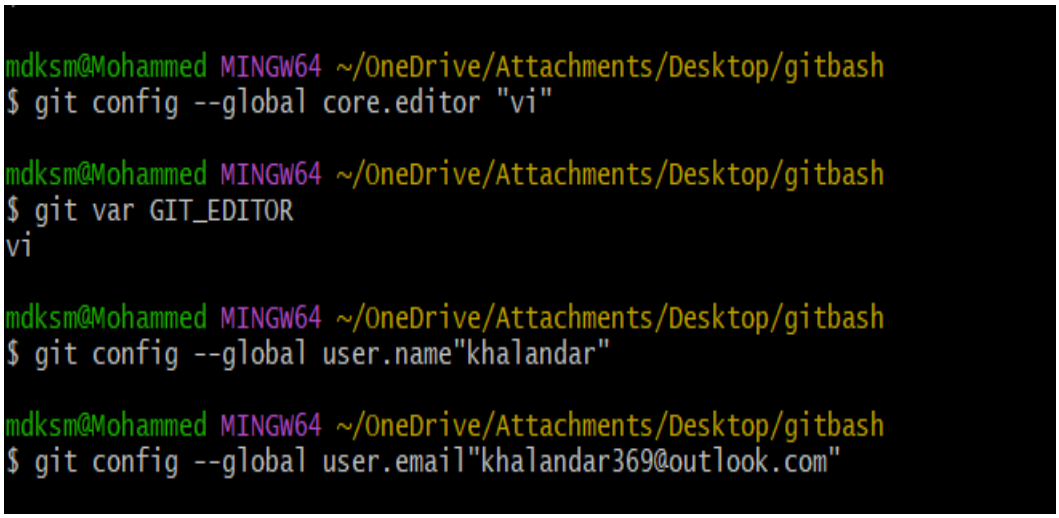
   Git git var GIT_EDITOR

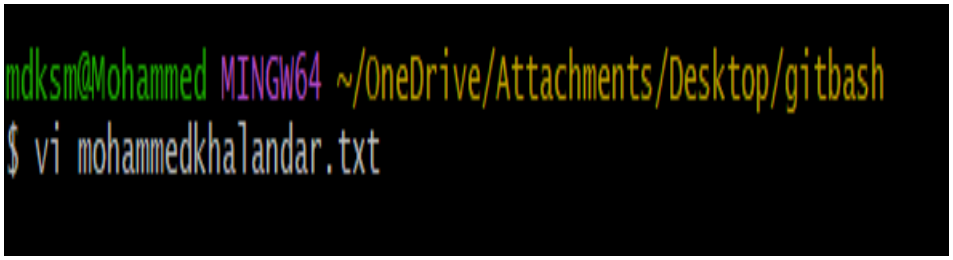
2. Set vi editor as the default editor for Git Bash

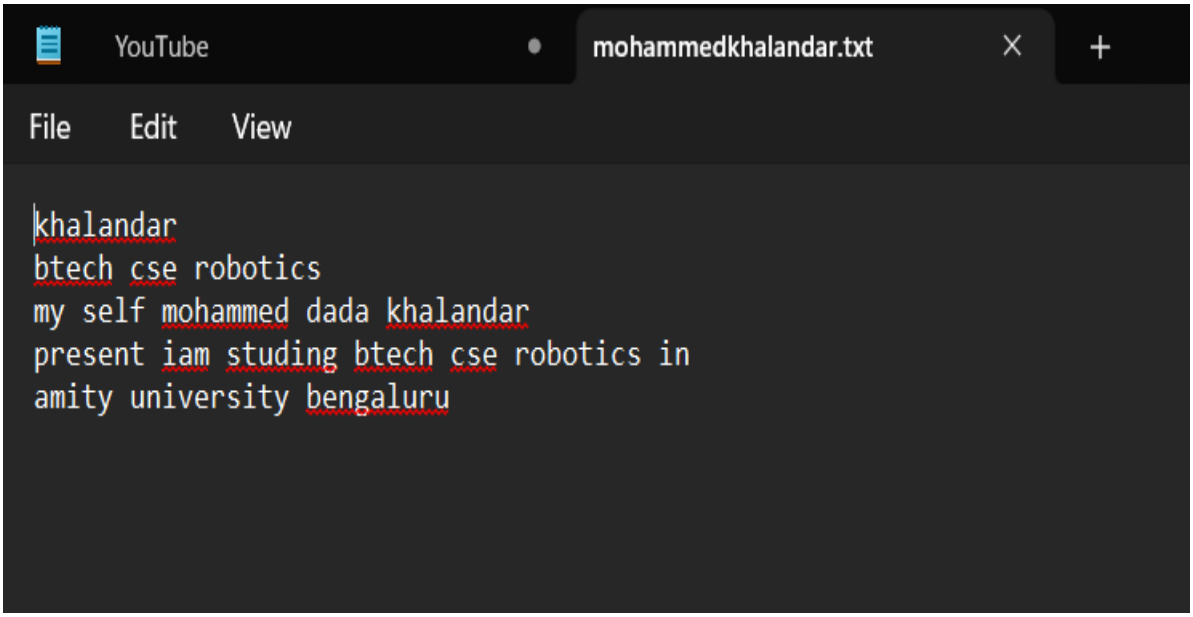
   git config --global core.editor "vi"

3. Opening a File

   vi filename.txt






i

Inserts text before the current cursor.

I

Inserts text at the beginning of line.

a

Insert text after the current cursor.

A

Insert text at the end of the current line.

o

Creates a new line for text entry below cursor location and switches to insert mode.

O

Creates a new line for text entry above cursor location and switches to insert mode.

:w

Save File

:q

Quit

:wq

Save and quit

:x

Save and quit

:q!

Quit without saving.

x

Delete character dd Delete current line

d$

Delete from cursor to end of line

d0

Delete from cursor to start of line

:q!

Quit without saving.

y

Delete character

p

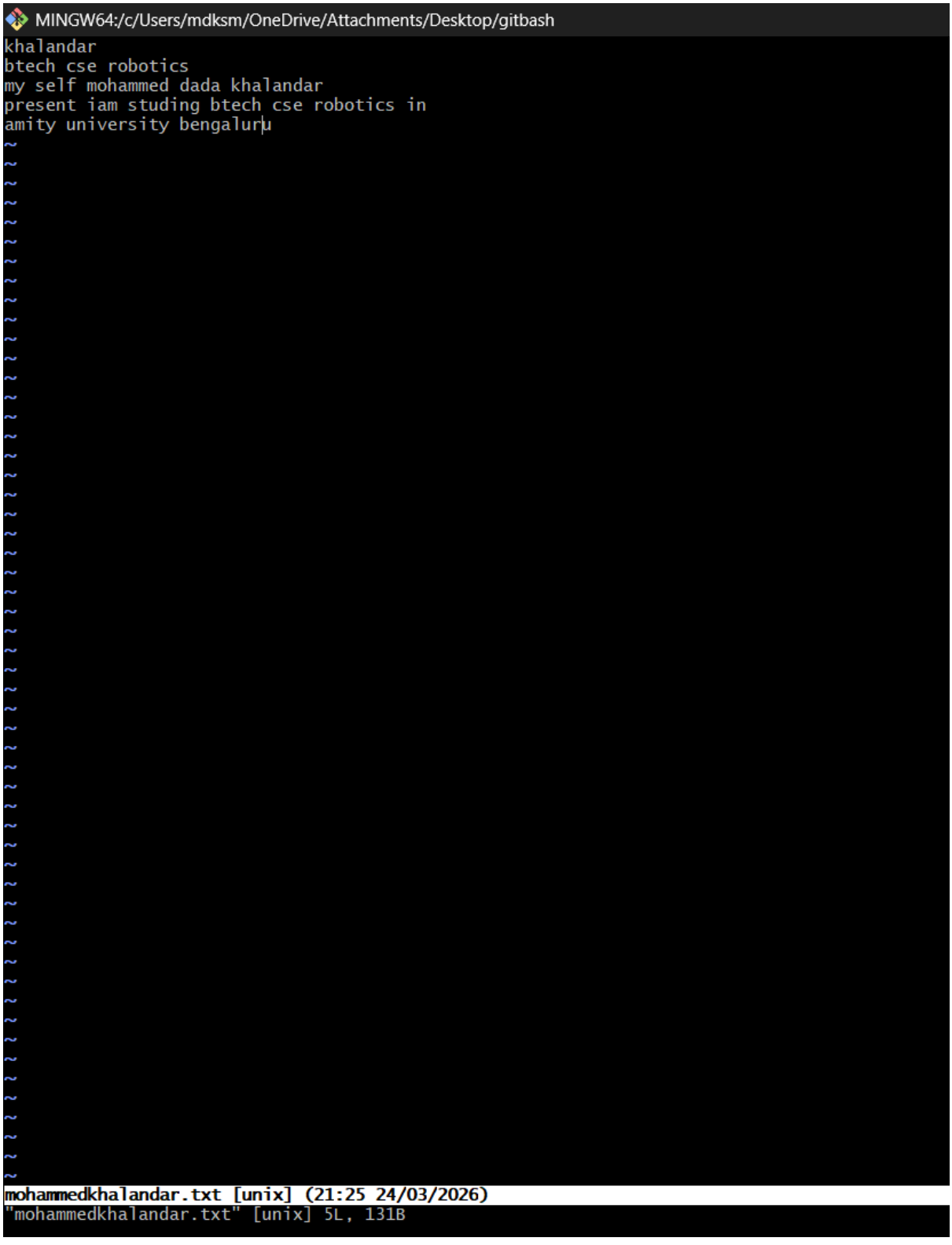
Delete current line

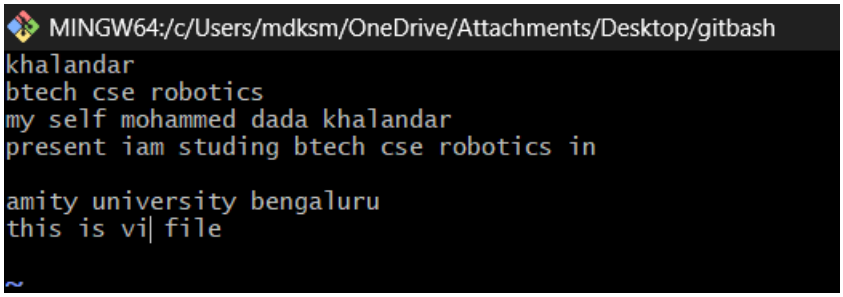
P

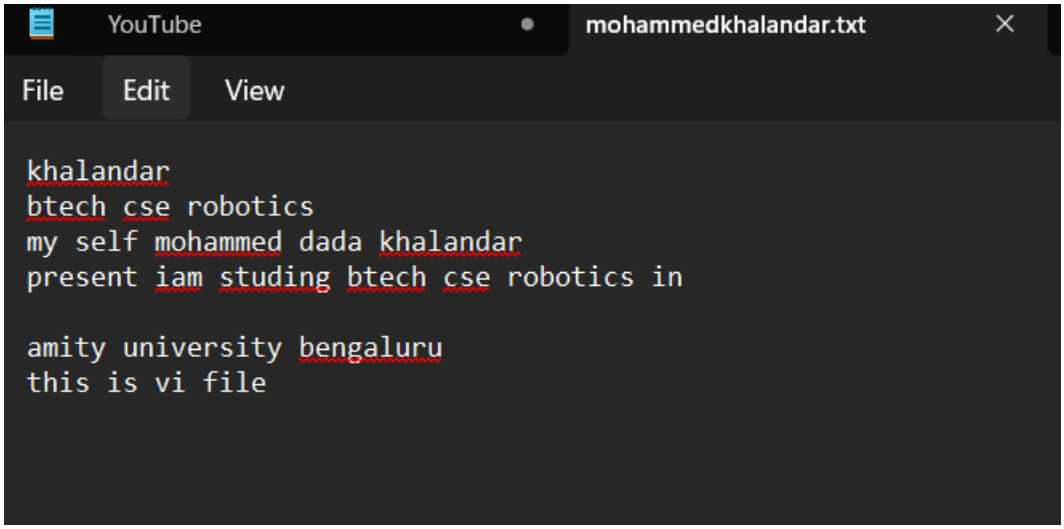
Delete from cursor to end of line

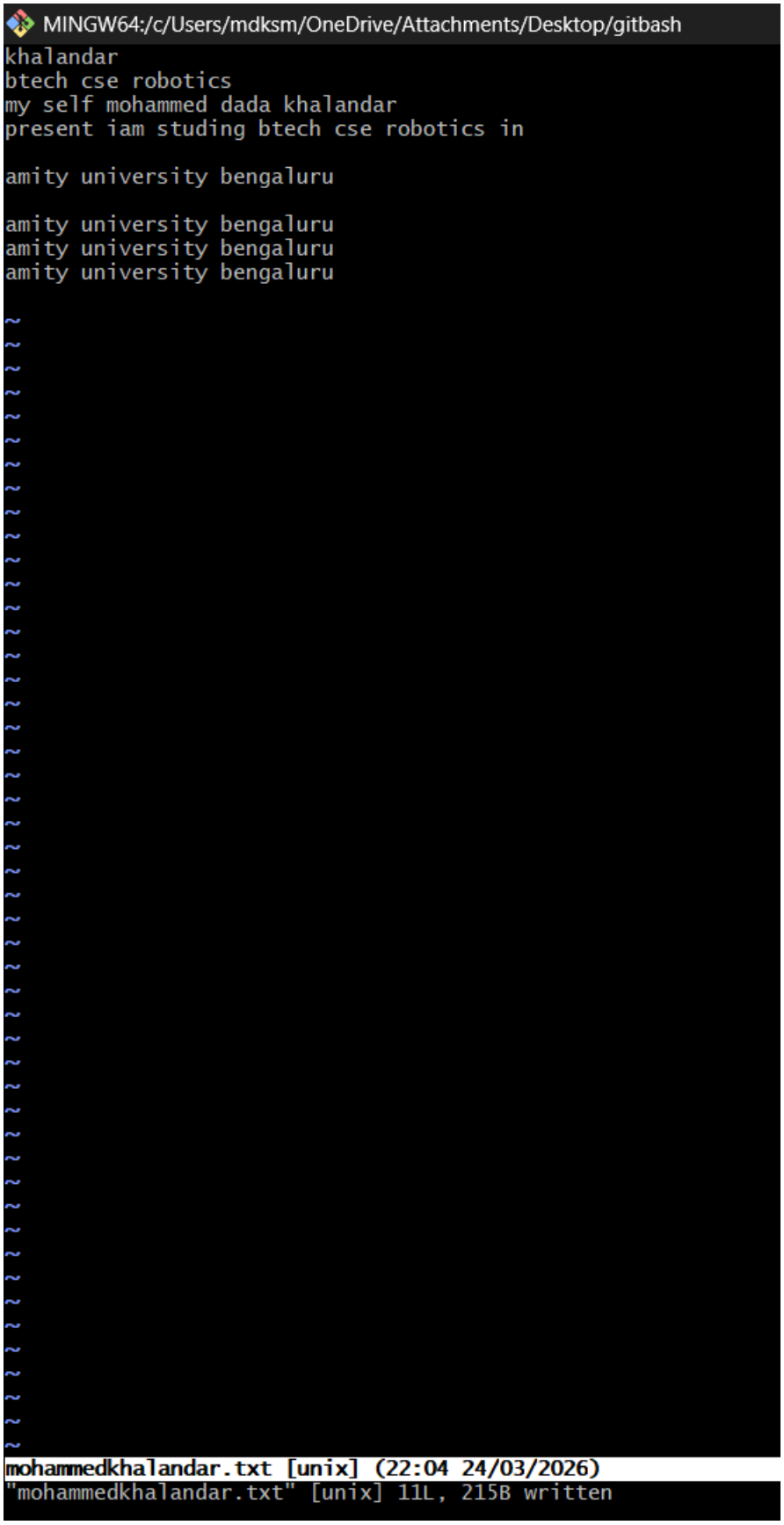
d0

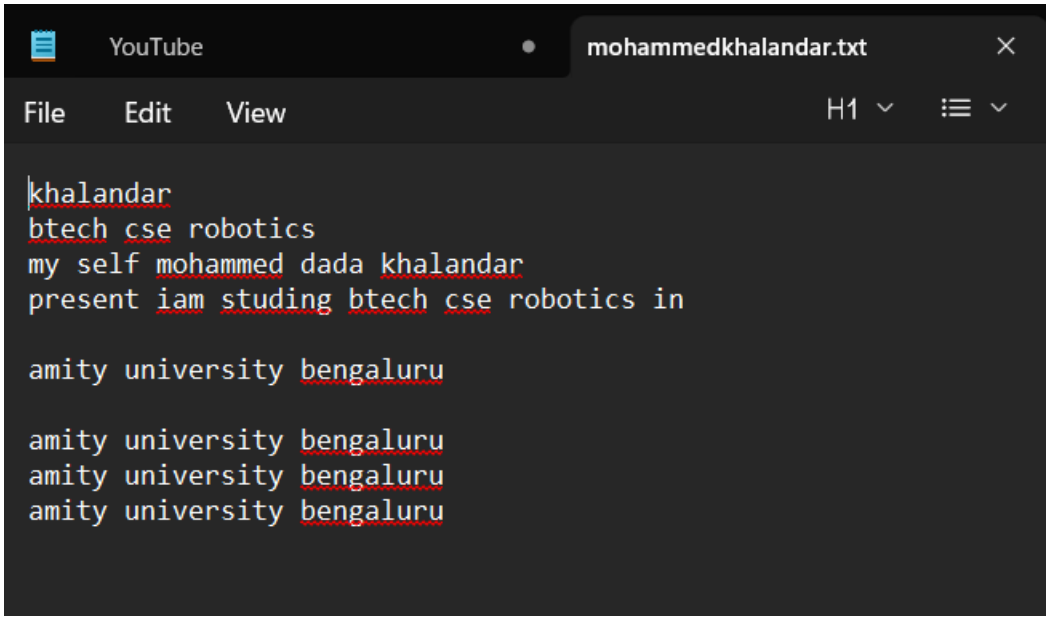
Delete from cursor to start of line


MINGW64:/c/Users/mdksm/OneDrive/Attachments/Desktop/gitbash
khalandar
btech cse robotics
my self mohammed dada khalandar
present iam studing btech cse robotics in
amity university bengaluru
mohammedkhalandar.txt [unix] (21:25 24/03/2026)
"mohammedkhalandar.txt" [unix] 5L, 131B


MINGW64:/c/Users/mdksm/OneDrive/Attachments/Desktop/gitbash
khalandar
btech cse robotics
my self mohammed dada khalandar
present iam studing btech cse robotics in
amity university bengaluru
this is vi file
~


YouTube
mohammedkhalandar.txt
File
Edit
View
khalandar
btech cse robotics
my self mohammed dada khalandar
present iam studing btech cse robotics in
amity university bengaluru
this is vi file


MINGW64:/c/Users/mdksm/OneDrive/Attachments/Desktop/gitbash
khalandar
btech cse robotics
my self mohammed dada khalandar
present iam studing btech cse robotics in
amity university bengaluru
amity university bengaluru
amity university bengaluru
amity university bengaluru
mohammedkhalandar.txt [unix] (22:04 24/03/2026)
"mohammedkhalandar.txt" [unix] 11L, 215B written


YouTube
mohammedkhalandar.txt
File
Edit
View
H1
khalandar
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